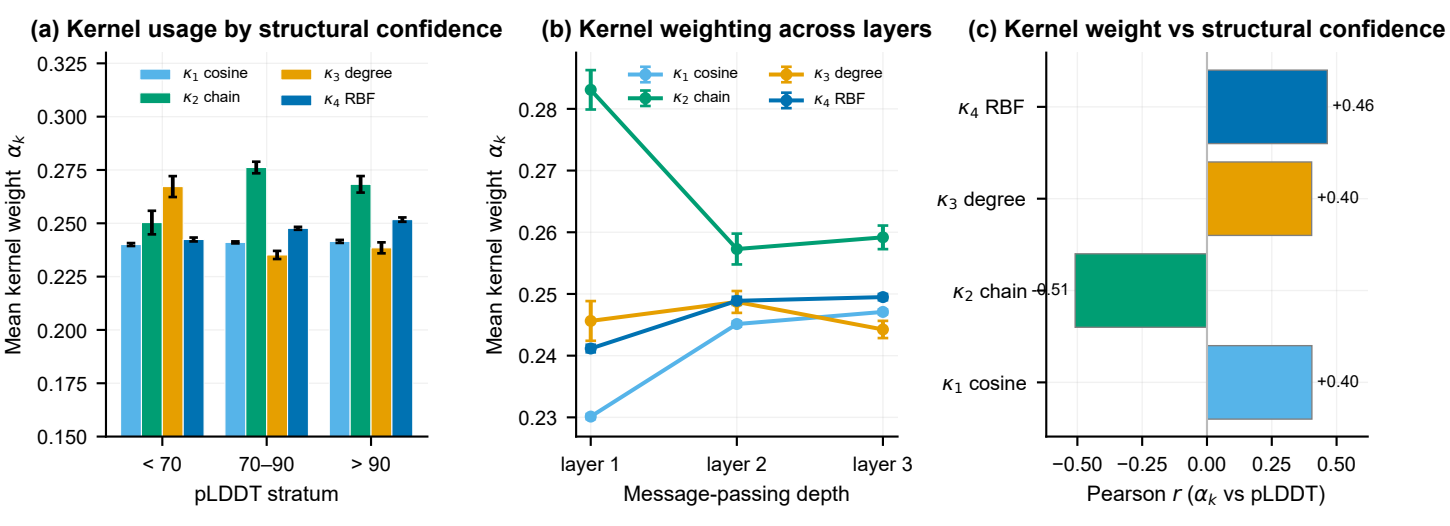

(a) Kernel usage by structural confidence
Mean kernel weight $\alpha_k$
0.325
0.300
0.275
0.250
0.225
0.200
0.175
0.150
$\kappa_1$ cosine
$\kappa_2$ chain
$\kappa_3$ degree
$\kappa_4$ RBF
< 70
70–90
> 90
pLDDT stratum
(b) Kernel weighting across layers
Mean kernel weight $\alpha_k$
0.28
0.27
0.26
0.25
0.24
0.23
$\kappa_1$ cosine
$\kappa_2$ chain
$\kappa_3$ degree
$\kappa_4$ RBF
layer 1
layer 2
layer 3
Message-passing depth
(c) Kernel weight vs structural confidence
$\kappa_4$ RBF
$\kappa_3$ degree
$\kappa_2$ chain
$\kappa_1$ cosine
+0.46
+0.40
−0.51
+0.40
−0.50
−0.25
0.00
0.25
0.50
Pearson $r$ ($\alpha_k$ vs pLDDT)